

8	(a) nucleus: <u>small</u> central part/core of an atom nucleon: proton or a neutron particle contained within a nucleus	B1 B1 B1	[3]
(b)	(i) 1. decay constant $= \ln 2 / (3.8 \times 24 \times 3600)$ $= 2.1 \times 10^{-6} \text{ s}^{-1}$	C1 A1	[2]
	2. $A = \lambda N$ $97 = 2.1 \times 10^{-6} \times N$ $N = 4.6 \times 10^7$	C1 A1	[2]
	(ii) 1.0 m^3 contains $(6.02 \times 10^{23}) / (2.5 \times 10^{-2})$ air molecules	C1	
	ratio $= (4.6 \times 10^7 \times 2.5 \times 10^{-2}) / (6.02 \times 10^{23})$ $= 1.9 \times 10^{-18}$	A1	[2]

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Section B